# Few-Shot Target Recognition of Remote Sensing Images Based on Causal Consistency and Domain-Sensitive Channel Regularization

Jie Lai, Jie Geng, *Member, IEEE*, Weichen Ma, and Wen Jiang, *Member, IEEE*

***Abstract*—Remote sensing target recognition is significantly impacted by domain shifts between the source domain (SD) and target domain (TD), especially under few-shot conditions, where models struggle to extract effective features. Current methods emphasize extracting features that are invariant across domains, yet they often neglect the influence of domain-sensitive structures within convolutional networks, which hinders the overall generalization performance of the model. To address this issue, a novel few-shot domain generalization method named causal consistency and domain-sensitive channel regularization is proposed for remote sensing target recognition. Within the proposed framework, causal consistency learning is introduced to capture key causal features of target categories, mitigating the interference of noncausal features during inference. In addition, a domain-sensitive channel regularization module is designed to identify and suppress convolutional kernels with unstable activations caused by domain shifts, thereby preserving the structural robustness of the network and enhancing cross-domain feature extraction. To further stabilize training, a structural bias attenuation loss is incorporated to mitigate the instability introduced by kernel suppression, ensuring the model maintains robust generalization across different domain environments. The experimental results confirm the effectiveness of the proposed method in suppressing the impact of domain-sensitive structures, improving domain-invariant feature extraction, and achieving superior recognition performance under domain shift conditions compared to existing approaches.**

***Index Terms*—Causal learning, domain shift, domain-sensitive channel regularization, few-shot learning (FSL), remote sensing target recognition.**

## I. Introduction

REMOTE sensing target recognition, a key task in intelligent remote sensing interpretation, aims to identify object categories and determine their spatial locations [1]. Optical sensors and synthetic aperture radar (SAR) are two primary modalities used in remote sensing, each offering distinct advantages [2], [3]. Optical sensors, which rely on visible light (VIS), capture high-resolution images and provide abundant spectral and textural details. However, their performance is easily influenced by atmospheric conditions such as cloud cover and low-light environments, which can obstruct or degrade the quality of the captured images [4]. SAR, in contrast, utilizes microwave imaging by continuously transmitting and receiving radar signals [5], enabling detailed observation of the Earth's surface at high resolution. Unlike optical sensors, SAR operates independently of illumination conditions and can penetrate clouds, ensuring reliable imaging in all-weather and day-and-night scenarios [6]. The capability makes SAR particularly advantageous for applications where optical sensors are limited. Both VIS and SAR imagery play a crucial role in remote sensing target recognition and have been widely applied in fields such as military reconnaissance, resource exploration, environmental monitoring, and urban planning [7].

Deep learning has emerged as a transformative approach for target recognition in remote sensing imagery in recent years [8], transitioning from traditional handcrafted feature extraction to data-driven end-to-end learning. Compared with conventional methods, deep learning-based detection algorithms autonomously learn hierarchical semantic features, improving recognition accuracy in complex backgrounds and demonstrating strong adaptability in multisource, multiscale scenarios. To enhance detection performance, Guo et al. [9] proposed CenterNet++, integrating feature refinement and pyramid fusion. Zhang et al. [10] introduced Quad-FPN, leveraging multiple feature pyramids to enhance SAR ship detection. Zhao et al. [11] developed an attention receptive pyramid Network, combining receptive field models with convolutional attention to improve multiscale detection. Jinchuan et al. [12] utilized the faster R-convolutional neural network (CNN) framework for ship detection in remote sensing images, integrating feature fusion to enhance target recognition. Lu et al. [13] introduced an attention-based feature fusion structure within the SSD framework to enhance detection by integrating multilevel features. Tian et al. [14] improved YOLOv5s by incorporating a bidirectional feature pyramid network for better multiscale feature fusion. Although these methods enhance detection accuracy, their reliance on large-scale annotated datasets limits effectiveness in high-cost or data-scarce scenarios, particularly under few-shot conditions. Few-shot learning (FSL) addresses this challenge by enabling efficient target recognition with limited samples, which is crucial in applications like remote sensing, where annotation is costly and data acquisition is challenging. As a result, improving detection performance under few-shot conditions has become a key research focus.

Few-shot target detection (FSTD) techniques are generally divided into two main paradigms: meta-learning (ML)

Received 4 February 2026; revised 1 April 2026; accepted 13 April 2026. Date of publication 16 April 2026; date of current version 23 April 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62571440. *(Corresponding author: Jie Geng.)*

The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: gengjie@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3684824

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

and transfer learning (TL). In ML-based detection, models typically integrate features from support images with query image regions of interest (RoIs) to perform classification and regression. Han et al. [15] introduced a query-adaptive detection method, leveraging a heterogeneous graph to model relationships between support class prototypes and query bounding box features. Zhang et al. [16] proposed support-guided proposal generation (SPG), which enhances RoI feature maps in query images using dynamic convolution. This method strengthens category-relevant proposals while filtering out background and mismatched candidates, improving bounding box precision for novel classes. In TL detection, Chen et al. [17] employed a pretraining-then-finetuning framework, integrating background suppression and knowledge transfer regularization to enhance the model's focus on foreground representations. Wang et al. [18] adopted a similar two-stage approach but froze the feature extraction and region proposal modules during fine-tuning, updating only the bounding box regression head to enhance model transferability. Liu et al. [19] tackled the challenges of remote sensing imagery in few-shot detection by proposing a progressive bounding box regression network, addressing issues with complex aspect ratios and target orientations. Recently, more studies have explored domain shift in few-shot detection. Fu et al. [20] characterized domain discrepancies based on interclass variance, undefined boundaries, and image style. They improved target feature alignment through instance-level adaptation and introduced a soft-referencing strategy under the Transformer framework [21] to synthesize domain variations, enhancing model generalization.

However, most FSTD methods are developed under the assumption of distributional alignment between training and testing data [22], [23], failing to fully address the domain shift problem in remote sensing images under varying imaging conditions [24]. In real-world applications, differences in sensor types, viewing angles, surface characteristics, and environmental factors introduce significant feature distribution discrepancies between the source domains (SDs) and target domains (TD) [25]. As a result, models trained solely on SD data struggle to maintain stable detection performance in the TD [26]. Domain generalization aims to enhance model generalization without access to TD data. It focuses on learning robust features from the SD during training, enabling adaptation to unseen domains. However, most existing domain generalization-based target recognition methods are designed for classification tasks, with limited research on object detection. Current approaches often adapt techniques from classification models and primarily focus on global feature alignment. Yet, under few-shot conditions, key object features tend to be weak and easily overwhelmed by background interference, leading to degraded detection performance. More critically, in cross-modal tasks, differences in imaging mechanisms across sensors introduce complex feature discrepancies between the SD and TD, further increasing the challenge of model generalization [27]. Recent studies have attempted to mitigate cross-modal discrepancies by modeling modality-specific characteristics or disentangling shared and private representations. Specifically, Wang et al. [28] incorporated radar scattering mechanisms to improve SAR–RGB image registration under cross-modal discrepancies. He et al. [29] proposed a multimodal fusion framework that disentangles shared and modality-specific features through iterative optimization.

In summary, effective few-shot domain generalization relies on learning representations that emphasize domain-invariant target characteristics rather than domain-specific appearance variations. This can be achieved by enhancing feature robustness and suppressing domain-sensitive factors during training, thereby improving generalization to unseen domains.

To cope with the challenges delineated above, a few-shot domain generalization method for target recognition is proposed that integrates causal consistency learning with domain-sensitive channel regularization. The goal is to enhance model generalization and cross-domain adaptability under few-shot conditions. First, causal consistency learning is introduced to capture the causal relationships of target categories by extracting intrinsic causal features, such as stable geometric structures and shape outlines in remote sensing imagery. Meanwhile, suppressing spurious correlations induced by noncausal factors, including cross-modal discrepancies and noise, leads to more discriminative feature representations. Second, a domain-sensitive channel regularization module is designed to actively identify and suppress convolutional kernels with unstable activations induced by domain shifts. By optimizing cross-domain feature distributions, the module improves the model's adaptability under varying imaging conditions. Finally, a structural bias regulation strategy is proposed to reduce instability caused by suppressing domain-sensitive structures during training. It optimizes feature learning and enhances model robustness in cross-domain tasks. The main contributions of this work can be outlined as follows.

1) A domain generalization framework based on causal consistency is proposed to enhance the robustness of few-shot target recognition. By incorporating causal consistency loss, the framework guides the model to focus on extracting causal features of target categories, while effectively suppressing noncausal factors that could interfere with the inference process, thereby improving the model's discriminative performance in few-shot settings.
2) A domain-sensitive channel regularization module is proposed, which identifies and suppresses convolutional kernels with unstable activations caused by domain shifts through analysis and selective removal, thereby preserving domain-robust structures and enabling the extraction of domain-invariant features.
3) A structural bias regulation strategy is introduced to improve the model's transferability across different domains. This strategy adjusts structural distributions during feature learning, reducing domain-sensitive influences during training. It ensures that the model maintains strong generalization capability even without TD data, thereby enhancing the reliability of cross-domain target recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

This article is structured as follows. Section II outlines the background and related work. Section III describes the proposed method. Section IV presents the experimental procedures and interprets results. Section V concludes this study.

## II. Related Works

### A. Few-Shot Learning

FSL aims to train models that can adapt to new categories or tasks using limited labeled samples. In remote sensing applications, FSL often relies on ML or TL to leverage shared knowledge from data-rich SDs. For example, recent studies [30], [31], [32] have utilized task-centric training, spatial feature augmentation, and pseudo-labeling to improve model adaptability with minimal support samples. In TL, Wang et al. [33] proposed a two-stage approach that transfers knowledge from optical images to SAR targets. Cheng et al. [34] also explored meta-TL, using an independent causal constraint to extract transferable causal factors, providing valuable insights into robust feature learning. Yang et al. [35] proposed a pretraining-based TL framework with a knowledge inheritance initializer and an adaptive length rescaling strategy to improve weight transfer and accelerate adaptation in FSTD. GFENet [36] incorporated model perturbation to mitigate overfitting in TDs.

Despite these advances, cross-modal FSL remains challenging when large modality discrepancies are present. Although recent studies have explored transferable representations and even introduced causal constraints, existing approaches primarily focus on improving feature transferability at the representation level. The structural characteristics of targets that remain stable across modalities are not explicitly emphasized during feature adaptation, which may lead to entanglement between domain-invariant semantics and modality-dependent variations. Consequently, domain-sensitive responses can still introduce structural bias into the learned representations. To address this issue, it is desirable to further exploit causal target features while regulating feature responses from a channel perspective, enabling more robust domain-invariant representation learning under limited supervision.

### B. Domain Generalization

Domain generalization seeks to develop models capable of performing well on unseen TD, without utilizing any TD data during the training phase. In the single-source setting, the objective is to enhance the model's generalization ability to novel test domains using training data derived exclusively from one SD. Compared to multi-SD generalization, this approach is more aligned with real-world applications where collecting data from multiple distinct domains is often impractical. Qiao et al. [37] introduced an adversarial training approach to generate hard samples and apply ML for data augmentation. Shankar et al. [38] utilized a Wasserstein-based constraint within the feature space to generate adversarial examples. Wu and Deng [39] designed a self-distillation mechanism that explicitly separates features common to all domains from those unique to specific domains. Wang et al. [40] designed a style augmentation module to generate complementary training samples. Fan et al. [41] proposed a generalized normalization method that learns to normalize and rescale statistical information.

Existing domain generalization methods operate primarily at the distribution or feature level, focusing on aligning feature spaces after feature extraction. However, under complex domain shifts, certain convolutional channels produce highly sensitive and unstable activations. Aligning features solely at the network's end fails to eliminate these internal structural noises, making the model prone to overfitting, especially under few-shot conditions. This limitation highlights the necessity of channel-level processing. By actively suppressing unstable convolutional kernels, channel-level intervention blocks the propagation of domain-sensitive noise to deep layers, fundamentally guaranteeing stable cross-domain feature extraction.

## III. Methodology

### A. Problem Formulation

In the context of few-shot domain generalization, let $X$ be the input space, $Y$ be the label space, and $\mathcal{D}$ be the domain space. Each domain $d \in \mathcal{D}$ is associated with a joint distribution $P_{XY}^{(d)}$. For $K$ distinct SD $\mathcal{D}_{source} = \{\mathcal{D}_k\}_{k=1}^{K}$, each SD has its own distribution $P_{XY}^{(k)}$, with $P_{XY}^{(k)} \neq P_{XY}^{(k')}$ for $k \neq k'$, causing distribution shifts between domains. We aim to build a model $f : X \rightarrow Y$ trained on limited annotated data across diverse SD, capable of generalizing to the TD $\mathcal{D}_{target}$ with reduced prediction error. Each task includes a support set and a query set, where the support set contains $N$ classes, each with $K$ labeled examples, denoted as $\mathcal{S} = \{S_i\}_{i=1}^{N\times K} = \{(x_i, y_i, d_i)\}_{i=1}^{N\times K}$, and the query set consists of $N \times q$ unlabeled samples, denoted as $\mathcal{Q} = \{Q_i\}_{i=1}^{N\times q} = \{(x_i, y_i, d_i)\}_{i=1}^{N\times q}$. While traditional FSL focuses on learning novel categories $\mathcal{T}_{novel}$ with a few labeled samples from a base set $\mathcal{T}_{base}$, the task involves handling both the few-shot setting and the cross-domain distribution shifts, where the model must generalize from multiple SD $\mathcal{D}_{source}$ to an unseen TD $\mathcal{D}_{target}$, all while addressing the scarcity of labeled samples and domain-sensitive variations. The challenge lies in learning effective representations that generalize across domains and handle both FSL and domain shifts simultaneously.

### B. Overview

The overall architecture of the proposed model is presented in Fig. 1, consisting of causal consistency processing and domain-sensitive channel regularization. This framework aims to improve FL by reducing domain shift and promoting better generalization through the learning of shared features across domains. First, the causal consistency module constrains the network from two perspectives: global causal knowledge consistency and local causal feature alignment. Since the mechanism for extracting causal features remains consistent across different few-shot tasks, a twin network is introduced to preserve historical knowledge and prevent catastrophic forgetting, achieved via exponential moving average updates of the primary network's weights. In addition, a causal consistency loss is designed to strengthen the association between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

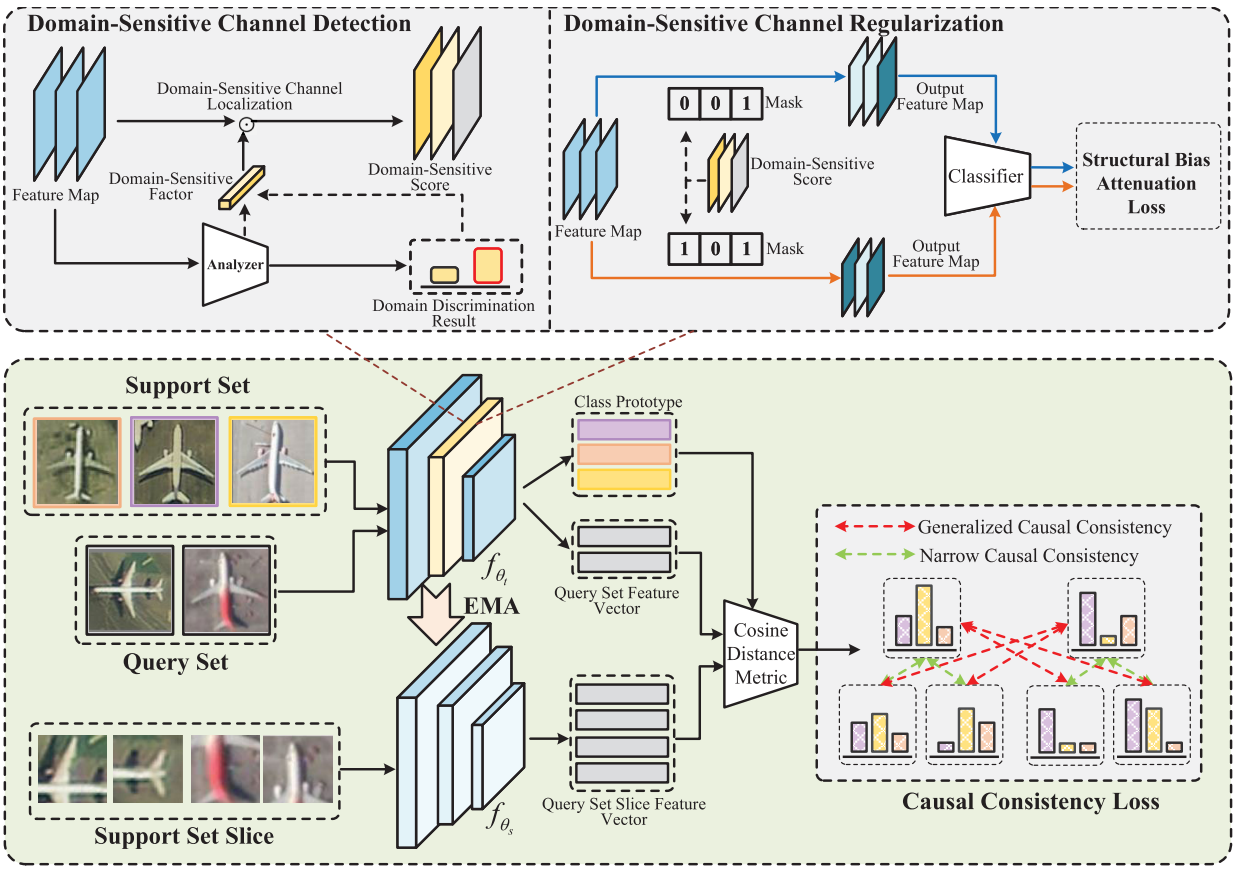



Fig. 1. Framework of the domain generalization few-shot target recognition method based on causal consistency and domain-sensitive channel regularization.

causal features and category prediction, effectively suppressing the influence of irrelevant or noncausal features. Then, the domain-sensitive channel regularization module adopts an analyze-then-filter strategy. A domain analyzer assigns sensitivity scores to each convolutional kernel based on its domain dependence. Kernels with high sensitivity scores, which indicate strong domain bias, are removed. After that, to maintain stable prediction under different filtering strategies, a structural bias attenuation loss is introduced to ensure consistency and robustness in cross-domain feature learning. The final model, trained with causal consistency learning and domain-sensitive channel regularization, is directly deployed for inference without requiring any parameter updates during testing.

### C. Causal Consistency Loss

Global causal knowledge learning aims to extract and preserve shared causal knowledge across tasks during meta-training, allowing the model to make correct decisions based on causal features. In this work, causal features are defined as intrinsic target properties, such as structural outlines and geometric shapes that remain invariant under domain shifts. Within our proposed framework, this concept is strictly operationalized as the representations extracted by the causal consistency module, explicitly disentangling core target structures from noncausal environmental background noise. In FSL, training involves tasks with randomly sampled categories, leading to different semantic spaces for each task. This forces the network to frequently switch between visual modes, making it difficult to accumulate stable causal knowledge. Despite these visual variations, the ability to infer a target's category based on causal features remains transferable, highlighting the importance of global causal knowledge learning. To address this issue, the proposed method introduces a twin network design, where the model parameters of the primary network are updated using exponentially moving averages. This process allows causal knowledge to be distilled across batches into the twin network, enabling it to better integrate global causal knowledge for local feature inference, significantly enhancing the generalization ability of the deep neural network. The update process is as follows:

$$\boldsymbol{\theta}_t \leftarrow m\boldsymbol{\theta}_t + (1-m)\,\boldsymbol{\theta}_s \tag{1}$$

where $\boldsymbol{\theta}_t$ represents the model weights of the primary network, $\boldsymbol{\theta}_s$ represents the model weights of the twin network, and $m$ denotes the coupling coefficient for weight updates.

In addition, global causal knowledge learning is updated gradually through exponential moving averages, and the introduction of incorrect causal knowledge can lead to biased gradient updates, affecting knowledge accumulation. Traditional target recognition models use deep CNNs to map images to high-level semantic spaces but fail to address interference features, causing irrelevant features to contaminate the semantic space and hinder target recognition. While

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

global semantic features aid in recognition, interference from noncausal features remains. The differences between target types are often distinguishable by local causal features, which enhance recognition performance. To address this challenge, a causal consistency loss is introduced to enhance the extraction of causal representations and support robust learning of global causal knowledge. In addition, a causal feature prediction branch is employed to measure the association between local features and their respective categories. If the local feature is causal, the correlation is strong; if noncausal, the correlation is weak. Training on ML tasks allows the model to accurately predict causal features in the query set, avoiding false associations caused by noncausal features.

Specifically, the global semantic features of each image are extracted using the primary network $f_{\theta_t}$, and the average semantic representation $c_i$ for each target category is computed according to (2). For the global semantic features $q_j$ of the query set, their correlation $c_i$ with the category representations $r_{q_j}$ is calculated using (3)

$$c_i = \frac{1}{K}\sum_{k=1}^{K} f_{\boldsymbol{\theta}_t}\left(X_{S_{i,k}}\right) \tag{2}$$

$$r_{q_j} = \text{match}\left(f_{\boldsymbol{\theta}_t}\left(X_{q_j}\right), c_i\right), \quad i \in [1, N] \tag{3}$$

$$r_{q_{j,r}} = \text{match}\left(f_{\boldsymbol{\theta}_s}\left(X_{q_{j,r}}\right), c_i\right), \quad i \in [1, N] \tag{4}$$

where $K$ represents the total number of samples in the support set for the $i$th category, $X_{S_{i,k}}$ represents the $i$th sample of the $k$th category in the support set, *match* denotes the cosine similarity calculation, and $X_{q_j}$ represents the $j$th sample in the query set.

In the local feature prediction branch, the features of randomly cropped slices from each query image, along with the category representations $c_i$ calculated using (2), are used to compute the correlation between the local slices and each category representation. Since local information in images can be effectively obtained through multiple slice augmentation, this process first performs random cropping on the original image and then rescales the cropped image patches to a lower resolution. This article uses multiple slice augmentation to obtain local slices for each query image. Specifically, for the $j$th query image $X_{q_j}$ in the ML task, its slices $X_{q_{j,r}}$, $r \in [1, R]$ are first obtained through multiple slice augmentation, where $R$ represents the total number of slices. Then, the subnetwork $f_{\boldsymbol{\theta}_s}$ extracts the features of these local slices $f_{\boldsymbol{\theta}_s}(X_{q_{j,r}})$. Similarly, the correlation $r_{q_{j,r}}$ between the local features and category representations is computed using (4). The causal consistency loss is calculated by measuring the consistency between the global semantic correlation $r_{q_j}$ and the local feature correlation $r_{q_{j,r}}$, and is then backpropagated during training. This enables the model to optimize global semantic features and suppress noncausal features. Specifically, for a query image and its local slices, the causal consistency loss is computed as follows:

$$\mathcal{L}^{\text{self}}_{X_{q_j}} = \frac{1}{R}\sum_{r=1}^{R} \mathcal{L}_{\text{CE}}\left(r_{q_{j,r}}, r_{q_j}\right) \tag{5}$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy loss. Due to the limitations of image resolution and multiple slice augmentation, the number of available positive and negative sample pairs for calculating the causal consistency loss within a single query image is too small. Therefore, we generalize the concept of positive and negative samples: the global semantic features and local causal features of all images within the same category are considered as positive pairs, while local causal features within the same category and local features from other categories are treated as negative pairs. This generalization significantly increases the number of available sample pairs, thus preventing high variance in the calculation of causal consistency loss due to the scarcity of sample pairs. The generalized causal consistency loss is computed as follows:

$$\mathcal{L}^{\text{cross}}_{X_{q_j}} = \frac{1}{R}\sum_{r=1}^{R} \mathcal{L}_{\text{CE}}\left(r_{q_{k,r}}, r_{q_j}\right), \quad k \neq j. \tag{6}$$

### D. Domain-Sensitive Channel Regularization Module

The cross-domain generalization ability of CNNs depends on the sensitivity of their convolutional kernels to domain differences. For instance, when a CNN is trained on VIS images as the SD and tested on SAR images as the TD, the model develops a fixed regularization pattern for the style features of VIS images. Specifically, when VIS images are input, the activation distribution of different convolutional kernels remains fixed. However, when the model is tested on SAR images, this fixed regularization pattern lacks robustness to domain differences, resulting in less discriminative features from the SAR images and a decline in target recognition accuracy in the SAR.

Let $D_S$ and $D_T$ be the SD and TD, respectively. The purpose of the domain-sensitive channel analysis module is to train a model $f(\cdot)$ using the SD $D_S$, during which domain-sensitive channels are identified and suppressed, allowing the model to maintain robust classification performance on the unseen TD $D_T$. This module guides the model to reduce the weight of domain-sensitive features during training. Inspired by the domain discriminator mechanism proposed in [42], a domain analyzer is introduced in the intermediate layers of the model. The analyzer consists of a global average pooling layer and a fully connected layer, which evaluates the contribution of each channel to domain prediction and identifies domain-sensitive redundant channels. For the input $x_i$ from the SD $D_S$ and its labels $y_i$, the features $f_l(x_i) \in \mathbb{R}^{C\times H\times W}$ are first extracted from the $l$th intermediate layer, where $\mathbf{C}$ denotes the channel dimension, and $\mathbf{H}$ and $\mathbf{W}$ correspond to the spatial dimensions of the feature map, respectively. The feature map $f_l(x_i)$ is then passed through the analyzer to predict the domain label and calculate the discriminative loss.

In the context of single-domain generalization explored in this study, since there is only one SD, multiple domains cannot be mixed during training to extract domain-invariant features. To construct an effective auxiliary domain, we introduce typical physical degradation mechanisms observed in real-world scenarios. Specifically, SAR imagery inherently contains multiplicative speckle noise, and applying Rayleigh-distributed speckle augmentation has been shown to help models learn robust, noise-invariant representations [43]. Similarly, in optical imaging, Rayleigh scattering can significantly weaken surface reflectance, thereby reducing the separability of target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

features [44]. These studies clearly indicate that statistical perturbations based on the Rayleigh distribution can effectively simulate distribution shifts caused by real-world environmental interferences. Therefore, in constructing auxiliary domains, we uniformly adopt a generalized Rayleigh distribution as the noise model. To construct a new domain independent of the SD during training, Rayleigh noise is added to the original SD images. Thus, the input and labels from the SD can be represented as $x_{i,k}$ and $y_{i,k}$, where $k = 0$ denotes the SD images, and $k = 1$ represents the new domain images. These two types of images are input into the domain analyzer to identify the channels most affected by the disturbance.

Specifically, to identify the channels containing domain-sensitive information, we assume that the channels with the greatest contribution to domain prediction likely contain such information. Each channel's association with domain-sensitive information is measured by assessing its influence on accurate domain classification. For the input $x_{i,k}$ and feature extractor $f_l$, the domain sensitivity score of the $j$th channel in the feature map $f_l(x_i)$ is defined as

$$s_j = W_j^k \cdot \text{Pooling}\left(f_l\left(x_{i,k}\right)\right)_j \tag{7}$$

where $W_j^k \in \mathbb{R}^C$ is the domain-sensitive factor, representing the weights of the fully connected layer of the domain analyzer for domain classification results, $\mathbf{C}$ is the number of channels, and Pooling$(\cdot)$ denotes the average pooling applied to the $j$th channel. A larger weighted activation indicates a stronger contribution of the channel to precise domain classification. Channels with high contributions are selected as domain-sensitive channels and are processed by the domain-sensitive channel regularization in the subsequent steps.

The design of the domain-sensitive channel regularization module aims to reduce domain-sensitive information in the feature map by selecting and discarding the most domain-sensitive channels during training. First, the probability $p_j$ of discarding the $j$th channel is calculated based on the domain sensitivity score $s_j$, and the calculation is as follows:

$$p_j = \frac{s_j}{\sum_{c=1}^{C} s_c}. \tag{8}$$

Finally, a binary mask $m \in \mathbb{R}^C$ is generated based on the discard probability of each channel, where $m_i$ has a probability $p_j$ of being set to 1, indicating that channels with relatively high domain sensitivity scores are more likely to be discarded.

### E. Structural Bias Attenuation Loss

In the domain-sensitive channel regularization module, dynamic channel dropping introduces stochastic structural perturbations into the network. While this mechanism is necessary for suppressing domain-specific features, it may lead to inconsistent predictions for the same input, thereby causing training instability. To address this issue, we introduce the structural bias attenuation loss, which explicitly serves as a stability regularizer. Promoting predictive consistency, it encourages the model to maintain stable predictions during domain-invariant representation learning, even under structural variations caused by stochastic channel removal. Specifically, for each training batch, two independent stochastic channel-dropping operations are applied to the same input image $x_i$, producing two prediction distributions, $\hat{f}(x_1)$ and $\hat{f}(x_2)$. The symmetric Kullback–Leibler (KL) divergence between these predictions is minimized to constrain prediction variance and stabilize optimization, thereby encouraging representations that are robust to domain-sensitive structural perturbations. The structural bias attenuation loss is computed as follows:

$$\mathcal{L}_{\text{cons}} = \frac{1}{2}\left(D_{KL}\left(\frac{\sigma\left(\hat{f}\left(x_1\right)\right)}{T} \middle\| \frac{\sigma\left(\hat{f}\left(x_2\right)\right)}{T}\right) + D_{KL}\left(\frac{\sigma\left(\hat{f}\left(x_2\right)\right)}{T} \middle\| \frac{\sigma\left(\hat{f}\left(x_1\right)\right)}{T}\right)\right) \tag{9}$$

where $\sigma$ represents the softmax function with a regularization coefficient $T$, and $D_{KL}$ denotes the KL divergence.

In summary, the final loss function of the method proposed in this article consists of the causal consistency loss, structural bias attenuation loss, and the basic classification cross-entropy loss. The calculation process is as follows:

$$\mathcal{L} = w_1 \mathcal{L}_{X_{q_j}}^{\text{self}} + w_2 \mathcal{L}_{X_{q_j}}^{\text{cross}} + w_3 \mathcal{L}_{\text{cons}} + \mathcal{L}_{\text{CE}} \tag{10}$$

where $X_{qj}$ represents the $j$th query image sample, $w_1$ and $w_2$ denote the coupling weights for the local and global causal consistency losses, $w_3$ represents the coupling weight for the structural bias attenuation loss, and $\mathcal{L}_{CE}$ denotes the cross-entropy loss.

## IV. Experiment

### A. Dataset Description

The FAIR-1M dataset [45] is a multiclass, multiscale, fine-grained dataset designed for high-resolution remote sensing image target recognition tasks. It includes five major categories: aircraft, ships, vehicles, tanks, and sites. These categories are further subdivided into 37 subcategories. For instance, the aircraft category encompasses various types of fighter jets, commercial airplanes, and more; the ship category includes cargo ships, cruise ships, sailboats, warships, and other types of vessels; the vehicle category is divided into subcategories, such as sedans, trucks, buses, and more.

The SAR-AIRcraft-1.0 dataset [46] includes 16 463 aircraft instances across seven categories: A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and others. The categories common to both the FAIR-1M and SAR-AIRcraft-1.0 datasets—A220, A320/321, A330, ARJ21, Boeing737, Boeing787, and others—were selected for evaluating recognition accuracy.

The VEDAI dataset [47] is a publicly accessible dataset created for detecting vehicles in remote sensing images. It is specifically tailored for target detection tasks in high-resolution aerial imagery and includes multimodal remote sensing data, i.e., imaging data from different modalities captured at the same location and time. The images are standardized to a size of 512 × 512 pixels, making them suitable for various detection algorithms. The primary target category in this dataset is vehicles, which are further divided into nine types: cars,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

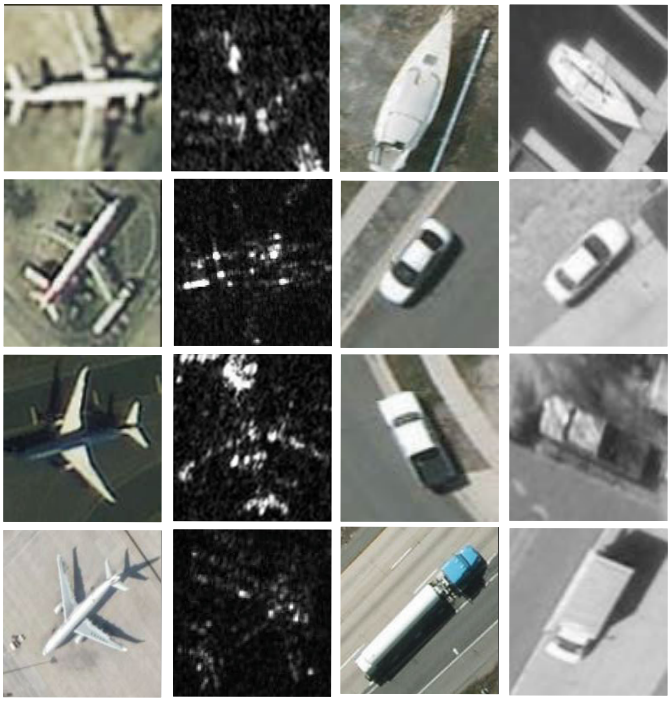



Fig. 2. Visualization of the datasets used in the experiments. (a) FAIR1M dataset, (b) SAR-AIRcraft-1.0, (c) VIS modality of the VEDAI dataset, and (d) IR modality of the VEDAI dataset.

trucks, vans, buses, trailers, semitrailers, agricultural vehicles, pickup trucks, and motorcycles.

The sample images from the datasets used in the experiments are shown in Fig. 2. The first column represents the FAIR1M dataset, the second column corresponds to the SAR-AIRcraft-1.0 dataset, and the third and fourth columns show the VIS and infrared (IR) images from the VEDAI dataset, respectively.

### B. Experimental Settings

The method proposed in this article was implemented using the PyTorch framework on a server with an NVIDIA GeForce RTX 4090 GPU. The model optimization utilized the Adam optimizer with a learning rate of 0.001, a weight decay of $5 \times 10^{-4}$, and default values for other parameters. To ensure fair comparisons, all reported experiments and baselines were trained for exactly 200 epochs, with an EMA momentum update coupling coefficient $m$ of 0.998. The weights for the local causal consistency loss $w_1$, global causal consistency loss $w_2$, and structural bias attenuation loss $w_3$ were set to 1.0, 0.15, and 0.001, respectively. The input image size was $84 \times 84$.

Regarding sample partitioning and few-shot task construction, this study adopts the standard episodic training paradigm tailored for cross-domain scenarios. Conventional approaches split a dataset into training and testing sets using a fixed ratio. In contrast, the data are partitioned by domain, where the SD is used exclusively for training while the TD is reserved entirely for testing. Under the standard five-way $K$-shot setting ($K = 1$ or 5), each few-shot episode is constructed by randomly sampling five categories. For each category, $K$ samples are selected as the support set, and 15 disjoint samples form the query set. During training, 100 episodes are sampled per epoch. For evaluation, the model is tested on 100 randomly sampled episodes from the unseen TD, and the average accuracy over these episodes is reported.

TABLE I
AVERAGE TARGET RECOGNITION ACCURACY ON THE **VIS** → **SAR** DOMAIN GENERALIZATION TASK

| Method | Trans* | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|
| RelationNet | | 20.02 ± 0.13 | 31.55 ± 0.54 |
| tSF | | 23.93 ± 0.26 | 26.14 ± 0.25 |
| GNN+FWT | ✓ | 22.10 ± 0.44 | 31.62 ± 0.56 |
| TPN+FWT | ✓ | 22.50 ± 0.47 | 23.75 ± 0.37 |
| GNN+ATA | ✓ | 22.20 ± 0.43 | 30.21 ± 0.53 |
| TPN+ATA | ✓ | 23.11 ± 0.43 | 27.63 ± 0.44 |
| TSA | | 23.47 ± 0.37 | 25.81 ± 0.25 |
| **Ours** | ✓ | **24.27 ± 0.25** | **33.07 ± 0.30** |

*Trans indicates the transductive setting.

To assess the performance of the proposed approach, we compare the model's recognition accuracy with other domain generalization target recognition algorithms in two common domain-shift remote sensing image interpretation scenarios: **VIS** (FAIR-1M) → **SAR** (SARAircraft-1.0) and **VIS** (VEDAI) → **IR** (VEDAI). Due to their abundant structural and texture information, VIS images facilitate the learning of generalizable features. Therefore, in both domain shift experiments, VIS images are utilized as the SD training data, and target recognition accuracy is tested on the SAR and IR images.

### C. Comparison With SOTA Methods

*1) Experiments With VIS as the SD and SAR as the TD:* This experiment uses the FAIR-1M dataset as the VIS SD and SAR-Aircraft-1.0 as the SAR TD to compare the accuracy of the proposed causal consistency and domain-sensitive channel regularization-based domain generalization method for few-shot target recognition. The methods compared include classic FSL approaches and recent advances that optimize FSL for domain shift. These methods are RelationNet [48], tSF [49], GNN + FWT [50], GNN + ATA [51], TPN + FWT [50], TPN + ATA [51], and TSA [52]. Among these, RelationNet and tSF are classic few-shot metric learning methods, where tSF further introduces a lightweight Transformer-based semantic filtering module to enhance task-relevant feature representation, while FWT, ATA, and TSA are recent advanced methods designed for domain generalization in FSL, focusing on feature transformation, task parameter fine-tuning, and model parameter fine-tuning, respectively. The experimental results are analyzed quantitatively in this section, as shown in Table I.

In traditional FSL methods, RelationNet achieves average accuracies of 20.02% ± 0.13% and 31.55% ± 0.54% on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AVERAGE TARGET RECOGNITION ACCURACY ON THE **VIS → IR** DOMAIN GENERALIZATION TASK

| Method | Trans* | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|
| RelationNet | | 26.76 ± 0.58 | 32.65 ± 0.54 |
| tSF | | 25.37 ± 0.52 | 34.89 ± 0.58 |
| GNN+FWT | ✓ | 20.09 ± 0.37 | 29.71 ± 0.55 |
| TPN+FWT | ✓ | 27.35 ± 0.58 | 37.03 ± 0.65 |
| GNN+ATA | ✓ | 20.90 ± 0.38 | 25.80 ± 0.49 |
| TPN+ATA | ✓ | 27.28 ± 0.63 | 39.57 ± 0.69 |
| TSA | | 33.28 ± 0.47 | 42.03 ± 0.44 |
| **Ours** | ✓ | **34.67 ± 0.50** | **45.25 ± 0.70** |

*Trans indicates the transductive setting.

the five-way one-shot and five-way five-shot tasks, respectively, while tSF attains 23.93% ± 0.26% and 26.14% ± 0.25%, suggesting that both metric-based and Transformer-based semantic filtering methods still struggle under the severe domain shift. The improved methods GNN + FWT and GNN + ATA reach 22.10% ± 0.44% and 22.20% ± 0.43% on the five-way one-shot task, and 31.62% ± 0.56% and 30.21% ± 0.53% on the five-shot task, showing some improvement, though the gain is limited. TPN + ATA achieves 23.11% ± 0.43% on the one-shot task and 27.63% ± 0.44% on the five-shot task, outperforming traditional methods but still lacking stability. The method proposed in this article achieves average accuracies of 24.27% ± 0.25% on the five-way one-shot task and 33.07% ± 0.30% on the five-way five-shot task, outperforming the comparison methods and demonstrating stronger cross-modal generalization ability. As the number of samples increases from one-shot to five-shot, the proposed method maintains steady performance growth with an 8.80% improvement, exhibiting superior performance compared to traditional methods like RelationNet and improved methods like the GNN and TPN series. It significantly improves few-shot target recognition accuracy under domain shift conditions by achieving the highest results in both the one-shot and five-shot tasks, effectively outperforming all comparison methods.

*2) Experiments With VIS as the SD and IR as the TD:* This experiment uses the FAIR-1M dataset as the VIS SD and VEDAI as the IR TD, comparing the performance of different FSL methods in domain generalization tasks. The experimental results are shown in Table II. In the five-way one-shot task, RelationNet achieves an average accuracy of 26.76% ± 0.58%, and in the five-way five-shot task, 32.65% ± 0.54%, while tSF attains 25.37% ± 0.52% and 34.89% ± 0.58% on the corresponding tasks, suggesting that Transformer-based semantic filtering brings only marginal improvements and both methods still exhibit limited performance in cross-modal scenarios. GNN + FWT and GNN + ATA achieve average accuracies of 20.09% ± 0.37% and 20.90% ± 0.38% in the one-shot task, respectively, and improve to 29.71% ± 0.55% and 25.80% ± 0.49% in the five-shot task, still showing suboptimal performance. TPN + FWT achieves accuracies of 27.35% ± 0.58% in the one-shot task and 37.03% ± 0.65% in the five-shot task, while TPN + ATA reaches 27.28% ± 0.63% and 39.57% ± 0.69%, demonstrating better domain generalization adaptability. The proposed approach attains an average accuracy of 34.67% ± 0.50% on the five-way one-shot task and 45.25% ± 0.70% on the five-way five-shot task. These results show that the proposed method exhibits strong generalization ability in domain generalization tasks, effectively handling feature transfer and adaptation between different modalities. In addition, the proposed method demonstrates stronger stability and learning capability as the number of samples increases, effectively extracting target semantic features from limited samples.

In addition, in the domain generalization task of VIS to IR, the average target recognition accuracy tends to be higher compared to the VIS to SAR cross-modal task. This phenomenon can be attributed to three main reasons. First, the similarity between VIS and IR modalities is higher. Second, there are significant differences between SAR and VIS images. SAR images, based on microwave imaging, have distinct imaging principles and characteristics compared to VIS. SAR images primarily capture structural information of targets, rather than texture and color. Thus, cross-modal transfer from VIS to SAR faces greater challenges, resulting in relatively lower recognition accuracy. Finally, there are spectral differences. The VIS and IR wavelengths are adjacent, and their spectral information is more closely related. As a result, the model struggles to effectively transfer features learned from VIS to SAR images, leading to lower accuracy.

The domain generalization target recognition method proposed in this article achieved excellent performance in the VIS to IR setting, demonstrating strong recognition ability in both one-shot and five-shot tasks. It achieved an average accuracy of 45.25 ± 0.70 in the five-shot task, outperforming most comparison methods, with the advantage becoming more pronounced as the sample size increased. The above experimental results indicate its ability to effectively address domain shift in few-shot environments, showcasing strong generalization capability and robustness.

### *D. Ablation Study*

To assess how well the proposed method performs, we conducted ablation studies focusing on the contributions of the causal consistency loss and the domain-sensitive channel regularization module to target recognition. The experiments were carried out under two domain shift conditions: **VIS → SAR** and **VIS → IR**, with dataset settings consistent with the comparison experiments.

*1) VIS → SAR:* According to the ablation results in Table III, when only the causal consistency module is added to the baseline, the average accuracies for the five-way one-shot and five-way five-shot tasks reach 23.29% ± 1.10% and 27.59% ± 0.59%, respectively. Compared to the baseline model, it represents an absolute performance improvement of 3.10% in the one-shot setting and 2.00% in the five-shot setting. When only the domain-sensitive channel regularization module is applied, the average accuracies

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDY RESULTS ON THE **VIS → SAR** DOMAIN GENERALIZATION TASK

| Index | Causal Consistency | Domain-Sensitive Channel Regularization | Structural Bias Attenuation | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|---|
| 1 | | | | 20.19±0.80 | 25.59±0.24 |
| 2 | ✓ | | | 23.29±1.10 | 27.59±0.59 |
| 3 | | ✓ | | 22.00±0.45 | 28.52±0.15 |
| 4 | ✓ | ✓ | | 22.77±1.12 | 30.95±1.32 |
| 5 | ✓ | ✓ | ✓ | **24.27±0.25** | **33.07±0.30** |

TABLE IV
ABLATION STUDY RESULTS ON THE **VIS → IR** DOMAIN GENERALIZATION TASK

| Index | Causal Consistency | Domain-Sensitive Channel Regularization | Structural Bias Attenuation | 5-way 1-shot | 5-way 5-shot |
|---|---|---|---|---|---|
| 1 | | | | 23.12 ± 0.38 | 31.60 ± 0.21 |
| 2 | ✓ | | | 25.73 ± 1.18 | 34.60 ± 0.33 |
| 3 | | ✓ | | 30.08 ± 0.69 | 37.49 ± 0.67 |
| 4 | ✓ | ✓ | | 33.89 ± 1.42 | 44.93 ± 1.70 |
| 5 | ✓ | ✓ | ✓ | **34.67 ± 0.50** | **45.25 ± 0.70** |

are 22.00% ± 0.45% and 28.52% ± 0.15%, respectively, yielding improvements of 1.81% and 2.93% over the baseline. A comparative analysis of these magnitudes reveals a clear distinction in their contributions. The causal consistency module provides a significantly larger magnitude of improvement in the extreme one-shot scenario, indicating its vital role in extracting structurally invariant causal features when data is severely limited. Conversely, the domain-sensitive channel regularization module contributes a greater performance gain in the five-shot setting, demonstrating its advantage in optimizing channelwise responses and suppressing domain-specific noise when provided with more support samples.

When both modules are combined, the average accuracy for the five-way one-shot task reaches 22.77% ± 1.12%, and the average accuracy for the five-way five-shot task is 30.95% ± 1.32%, both of which outperform the results from using each module individually. This result indicates that the combination of the causal consistency and domain-sensitive channel regularization modules effectively enhances the model's cross-modal generalization ability, with complementary effects between the two. The causal consistency module ensures that the model retains the ability to perform semantic reasoning based on local features, while the domain-sensitive channel regularization module effectively reduces the standard deviation of multiple test results, indicating that it improves the model's consistency in responding to various types of targets in the unseen domain during cross-domain tasks. After applying the structural bias attenuation loss, the model achieves average accuracies of 24.27% ± 0.25% on the five-way one-shot task and 33.07% ± 0.30% on the five-way five-shot task, indicating that this loss enhances the model's robustness to domain-sensitive channel regularization and facilitates the learning of domain-invariant features.

*2) Vis → IR:* According to the ablation results in Table IV, when only the causal consistency module is added to the baseline, the average accuracies for the five-way one-shot and five-way five-shot tasks reach 25.73% ± 1.18% and 34.60% ± 0.33%, respectively. Compared to the baseline model, it represents an accuracy improvement of 2.61% in the one-shot setting and 3.00% in the five-shot setting. Because the imaging characteristics of VIS and IR modalities share structural similarities, this module effectively preserves the consistency of label inference knowledge. When only the domain-sensitive channel regularization module is applied, the average accuracies rise to 30.08% ± 0.69% and 37.49% ± 0.67%, respectively, yielding substantial improvements of 6.96% in the one-shot setting and 5.89% in the five-shot setting over the baseline.

The above results indicate that in the VIS to IR domain generalization task, the domain-sensitive channel regularization module significantly outperforms the causal consistency module. This is mainly because RGB and IR images exhibit more similar imaging characteristics, whereas SAR imagery differs significantly from RGB imagery, making cross-modal transfer from optical to infrared easier. Consequently, extracting cross-domain invariant features by optimizing channel-wise responses becomes highly effective. Furthermore, when both modules are utilized simultaneously, the average accuracy improves to 33.89% ± 1.42% for the one-shot task and 44.93% ± 1.70% for the five-shot task. Finally, after applying the structural bias attenuation loss, the model achieves average accuracies of 34.67% ± 0.50% and 45.25% ± 0.70%, respectively, reaching peak performance. Meanwhile, the significant reduction in standard deviation demonstrates that this loss function effectively enhances the stability of the learned features and reduces sensitivity to modality-specific noise, thereby ensuring consistent predictions across different testing scenarios.

### *E. Hyperparameter Analysis*

To evaluate the impact of the momentum coefficient on model performance, Fig. 3 presents the ablation results on the **VIS → IR** task. The momentum coefficient plays an important role in regulating the update dynamics of model parameters, which is particularly critical under cross-modality transfer and data-scarce conditions. Following the classic Momentum Contrast mechanism proposed by He et al. [53], we adopt $m = 0.9$ as the starting value for our analysis.

1) When the momentum coefficient is set to $m = 0.9$, the parameter update is largely influenced by the current source information. Although this setting enables rapid adaptation to the support set, it is more susceptible to noisy gradients caused by extremely limited samples, resulting in unstable optimization behavior and relatively large performance variations across different runs.
2) When $m = 0.99$ is adopted, the update process benefits from increased historical accumulation, which partially alleviates instability in the early stages of training.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

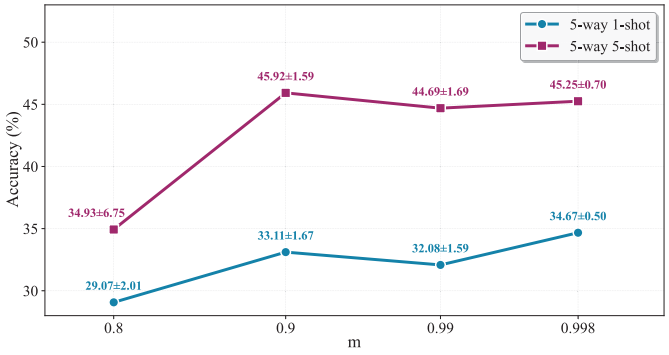



Fig. 3. Ablation study on the momentum coefficient $m$, showing the effect of different momentum settings on the convergence stability and performance variance.

However, the model remains sensitive to episodic gradient variations, and the accumulated information may interfere with effective adaptation to the current task distribution, leading to limited performance improvement and noticeable variance.

3) When a larger momentum coefficient $m = 0.998$ is employed, the update trajectory becomes substantially smoother. The stronger reliance on historical representations effectively suppresses sample-level noise, yielding more consistent convergence behavior and a significantly reduced performance variance across multiple runs. This property is especially desirable for the VIS-to-IR few-shot scenario, where robustness and reliability are of great importance.
4) When the momentum coefficient is set to $m = 0.8$, the recognition accuracy drops drastically, yielding only 29.07% and 34.93% in the one-shot and five-shot settings, respectively.

This ablation study highlights the importance of momentum control in few-shot recognition, demonstrating that an appropriate momentum setting not only improves accuracy but also enhances robustness and reliability.

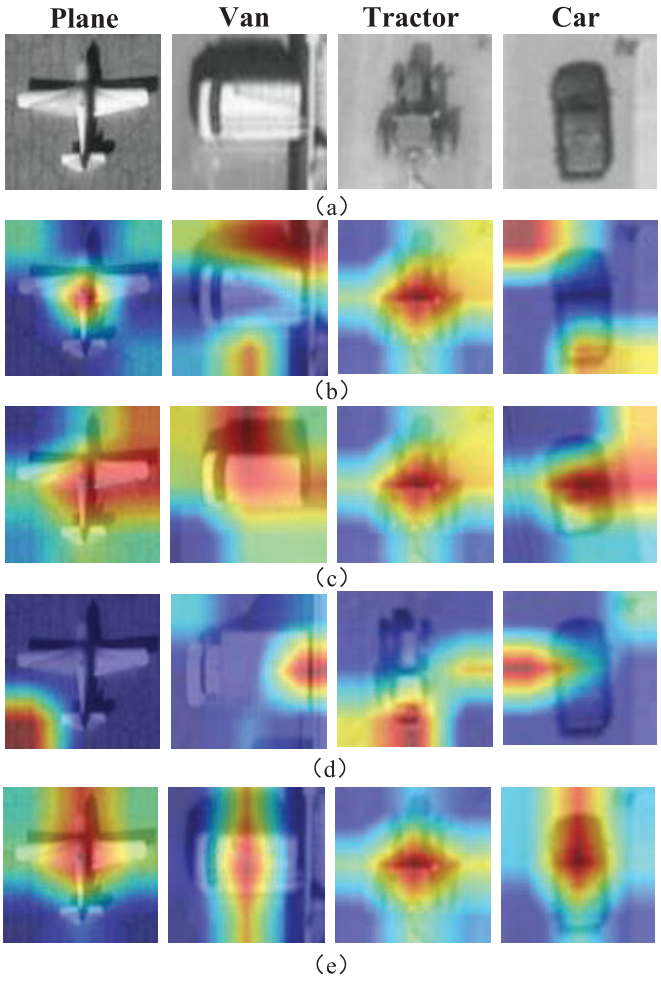



Fig. 4. LayerCAM visualization results of the proposed method in the five-way one-shot task under the VIS → IR conditional domain shift are shown. In the visualization results, key areas are shown in red to indicate high relevance, while blue denotes regions of lower importance. (a) Original input images. (b) Heatmaps generated by applying the causal consistency module to the model. (c) Heatmaps obtained using the domain-sensitive regularization module. (d) Heatmaps resulting from the combined effects of both modules in our method. (e) Heatmaps resulting from the combined effects of all three proposed modules.

### F. Visualization Analyses

To intuitively validate the effectiveness of the proposed method, Fig. 4 visualizes the feature heatmaps of different experimental groups in Table IV using LayerCAM [54]. We specifically selected the plane as the subject of this study, as its distinct geometric structural characteristics effectively test the model's ability to extract domain-invariant features. Observing the evolution of the heatmaps reveals that when only a single module is used, the heatmaps primarily exhibit a coarse, blob-like distribution. High-activation regions are confined to the geometric center of the target, failing to effectively cover key structural components such as the wings, while VIS diffusive activation persists in the background. These results suggest that the model's ability to capture the holistic structure of the target is insufficient at this stage. When the causal consistency and domain-sensitive channel regularization modules work synergistically, background interference noise is significantly suppressed, and the overall signal-to-noise ratio of the feature maps is markedly improved, laying a foundation for extracting pure target features. Finally, after introducing the structural bias attenuation loss, the heatmaps demonstrate clear structural adaptability. The activation regions are no longer shifted or scattered blobs but are tightly distributed along the fuselage and wing skeleton of the plane, presenting a topological morphology consistent with the physical structure. This transformation demonstrates that the proposed method effectively strips away modality-specific interference and forces the model to focus on stable global geometric features, thereby enhancing cross-modal domain generalization capabilities.

In addition, to validate the stable effect of the domain-sensitive channel regularization module on the network structure during cross-domain tasks, this experiment visualizes the activation levels of the 2nd-layer convolutional kernels in the ResNet-12 backbone network on TD. The experiment randomly sampled 500 images from the TD and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

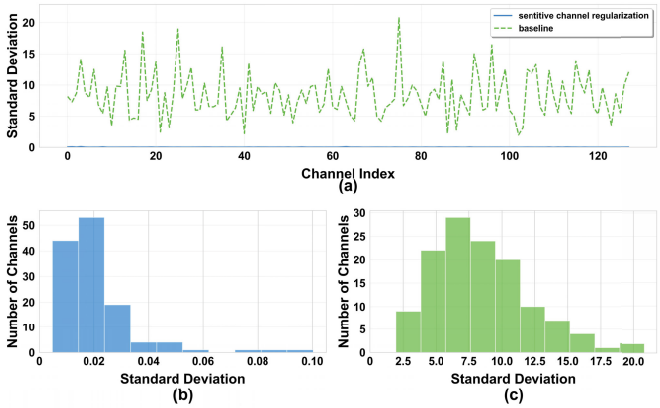



Fig. 5. Visualization of the standard deviation of activation values of the 2nd-layer convolutional kernels in the feature extraction network ResNet-12. (a) Standard deviation of activation values across channels. (b) Standard deviation distribution of activation values with domain-sensitive channel regularization. (c) Standard deviation distribution of activation values in the baseline model.

computed the standard deviation of the activation values of the convolutional kernels on these images. A higher standard deviation indicates unstable responses of the convolutional kernels in the TD, which may lead to feature distribution shifts in the model. In Fig. 5, the green curve represents the standard deviation of the activation values of the convolutional kernels at each layer in the model using only the causal consistency loss, while the blue curve represents the standard deviation in the model using both the causal consistency loss and the domain-sensitive channel regularization module. Specifically, Fig. 5(a) compares the green and blue curves within the same coordinate system, visually demonstrating that after employing the domain-sensitive channel regularization module, the model's activation responses become more stable. Fig. 5(b) and (c) shows the distribution of standard deviations along each colored curve, respectively, where each bar represents the number of channels whose activation values fall within a specific standard deviation range. By comparing these histograms, it is evident that the network with the domain-sensitive channel regularization module exhibits an activation distribution that tends toward lower standard deviation regions, where the majority of channels are concentrated, whereas the model without the module, represented by the green curve, shows that most channels' activation values are concentrated in higher standard deviation regions.

Overall, in the VIS to SAR and VIS to IR domain generalization tasks, the causal consistency module effectively reduces domain shift and enhances model performance. However, relying solely on the causal consistency module still fails to achieve optimal performance, especially in the VIS to SAR task, where modality differences are more significant. In contrast, the channel regularization module plays a crucial role in extracting domain-sensitive features, particularly in the VIS to IR task, where modality differences are smaller, significantly improving the model's cross-domain recognition ability. When both modules are combined, the synergy between the causal consistency module and the domain-sensitive channel regularization module further enhances the model's generalization ability and stability. The standard deviation analysis confirms that the domain-sensitive channel regularization module reduces the response fluctuations of convolutional layers in the TD, thereby improving the model's robustness. It demonstrates that the causal consistency module and the domain-sensitive channel regularization module are complementary in cross-domain target recognition tasks. Their combination effectively enhances model performance, particularly when addressing challenges between different modalities, showcasing stronger feature extraction and domain transfer capabilities.

## V. Conclusion

This work introduces an innovative few-shot target recognition approach for domain generalization, which incorporates causal consistency and domain-sensitive channel regularization to tackle the challenge of robustly capturing domain-invariant representations under domain shifts. Empirical results indicate that the proposed method delivers superior recognition performance across domains without the need for query set samples during inference, surpassing existing methods that rely on such information. The proposed method leverages causal consistency to ensure robust feature extraction and mitigate irrelevant factors. It also introduces a domain-sensitive channel regularization mechanism to identify and suppress domain-sensitive structures, enhancing the model's adaptability. A structural bias attenuation loss is incorporated to address training instability and refine feature extraction, ensuring stable performance across domains.

Overall, our work primarily focuses on improving few-shot target recognition under domain shift by incorporating causal intervention and enhancing feature robustness. However, it does not fully address the challenge of small target detection, particularly in remote sensing, where small targets occupy fewer pixels in the image and are difficult to distinguish from the background. Future research will focus on modeling the interaction between small targets and their background to improve recognition accuracy.

## References

[1] X. Yang, D. Kong, D. Yang, and M. Wang, "Domain-aware generalized meta-learning for space target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5638212.

[2] Z. Zhou, Z. Cui, Y. Tian, Y. Chen, Y. Pi, and Z. Cao, "Dynamic semantics-guided meta-transfer learning for few-shot SAR target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5209517.

[3] Z. Han et al., "Multisource collaborative domain generalization for cross-scene remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535815.

[4] Y. Bai et al., "IBFusion: An infrared and visible image fusion method based on infrared target mask and bimodal feature extraction strategy," *IEEE Trans. Multimedia*, vol. 26, pp. 10610–10622, 2024.

[5] B. Huang, T. Zhang, S. Quan, W. Wang, W. Guo, and Z. Zhang, "Scattering enhancement and feature fusion network for aircraft detection in SAR images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1936–1950, Feb. 2025.

[6] J. Geng, L. Dong, Y. Zhang, and W. Jiang, "Masked autoencoding and scatter-decoupling transformer for polarimetric SAR image classification," *Pattern Recognit.*, vol. 166, Oct. 2025, Art. no. 111660.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.

[7] W. Han, W. Jiang, J. Geng, and W. Miao, "Difference-complementary learning and label reassignment for multimodal semi-supervised semantic segmentation of remote sensing images," *IEEE Trans. Image Process.*, vol. 34, pp. 566–580, 2025.

[8] F. Lin, S. Ge, K. Bao, C. Yan, and D. Zeng, "Learning shape-biased representations for infrared small target detection," *IEEE Trans. Multimedia*, vol. 26, pp. 4681–4692, 2024.

[9] H. Guo, X. Yang, N. Wang, and X. Gao, "A CenterNet++ model for ship detection in SAR images," *Pattern Recognit.*, vol. 112, Apr. 2021, Art. no. 107787.

[10] T. Zhang, X. Zhang, and X. Ke, "Quad-FPN: A novel quad feature pyramid network for SAR ship detection," *Remote Sens.*, vol. 13, no. 14, p. 2771, Jul. 2021.

[11] Y. Zhao, L. Zhao, B. Xiong, and G. Kuang, "Attention receptive pyramid network for ship detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2738–2756, 2020.

[12] W. Jinchuan et al., "Research on remote sensing image target recognition method based on faster r-CNN deep network," *J. Earth Inf. Sci.*, vol. 20, no. 10, pp. 1500–1508, 2018.

[13] X. Lu, J. Ji, Z. Xing, and Q. Miao, "Attention and feature fusion SSD for remote sensing object detection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[14] Z. Tian, J. Huang, Y. Yang, and W. Nie, "KCFS-YOLOv5: A high-precision detection method for object detection in aerial remote sensing images," *Appl. Sci.*, vol. 13, no. 1, p. 649, Jan. 2023.

[15] G. Han, Y. He, S. Huang, J. Ma, and S.-F. Chang, "Query adaptive few-shot object detection with heterogeneous graph convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3243–3252.

[16] L. Zhang, S. Zhou, J. Guan, and J. Zhang, "Accurate few-shot object detection with support-query mutual guidance and hybrid loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14424–14432.

[17] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 2836–2843.

[18] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[19] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.

[20] Y. Fu et al., "Cross-domain few-shot object detection via enhanced open-set object detector," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 247–264.

[21] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[22] J. Geng, X. Deng, X. Ma, and W. Jiang, "Transfer learning for SAR image classification via deep joint distribution adaptation networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5377–5392, Aug. 2020.

[23] J. Geng, W. Ma, and W. Jiang, "Causal intervention and parameter-free reasoning for few-shot SAR target recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12702–12714, Dec. 2024.

[24] Z. Xu, W. Jiang, and J. Geng, "Dual-branch dynamic modulation network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507505.

[25] Y. Jiao, H. Yao, B.-K. Bao, and C. Xu, "Source-guided target feature reconstruction for cross-domain classification and detection," *IEEE Trans. Image Process.*, vol. 33, pp. 2808–2822, 2024.

[26] J. Geng, S. Song, and W. Jiang, "Dual-path feature aware network for remote sensing image semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3674–3686, May 2024.

[27] L. Dong, W. Jiang, Z. Xu, and J. Geng, "Multimodal cross-city semantic segmentation based on similarity-inspired fusion and invertible transformation learning network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 37, no. 3, pp. 1206–1220, Mar. 2026.

[28] P. Wang et al., "CIRSM-Net: A cyclic registration network for SAR and optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610619.

[29] Z. He et al., "DADIGAN: A dual attention blocks-based disentangled iterative generative adversarial network for cloud and shadow removal on SAR and optical images," *Inf. Fusion*, vol. 125, Jan. 2026, Art. no. 103487. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1566253525005603

[30] H. Luo, N. Jiang, H. Wang, J. Guo, and J. Zhu, "Meta-learning classification network for few-shot polarimetric SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[31] X. Dong, T. Ouyang, S. Liao, B. Du, and L. Shao, "Pseudo-labeling based practical semi-supervised meta-training for few-shot learning," *IEEE Trans. Image Process.*, vol. 33, pp. 5663–5675, 2024.

[32] H. Ji et al., "Semi-supervised few-shot classification with multitask learning and iterative label correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624315.

[33] X. Wang, H. Zhou, Z. Chen, J. Bai, J. Ren, and J. Shi, "Few-shot SAR ship image detection using two-stage cross-domain transfer learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 2195–2198.

[34] Y. Cheng, W. Zhang, H. Wang, and X. Wang, "Causal meta-transfer learning for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521014.

[35] Z. Yang, C. Zhang, R. Li, Y. Xu, and G. Lin, "Efficient few-shot object detection via knowledge inheritance," *IEEE Trans. Image Process.*, vol. 32, pp. 321–334, 2023.

[36] X. Ke, Q. Chen, H. Liu, and W. Guo, "GFENet: Generalization feature extraction network for few-shot object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 12741–12755, Dec. 2024.

[37] F. Qiao, L. Zhao, and X. Peng, "Learning to learn single domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12553–12562.

[38] S. Shankar, V. Piratla, S. Chakrabarti, S. Chaudhuri, P. Jyothi, and S. Sarawagi, "Generalizing across domains via cross-gradient training," 2018, *arXiv:1804.10745*.

[39] A. Wu and C. Deng, "Single-domain generalized object detection in urban scene via cyclic-disentangled self-distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 847–856.

[40] Z. Wang, Y. Luo, R. Qiu, Z. Huang, and M. Baktashmotlagh, "Learning to diversify for single domain generalization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 814–823.

[41] X. Fan, Q. Wang, J. Ke, F. Yang, B. Gong, and M. Zhou, "Adversarially adaptive normalization for single domain generalization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8208–8217.

[42] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.

[43] A. Rahimi and S. X. Yu, "Noise-tolerant novel-view SAR synthesis via denoising diffusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5202111.

[44] H. Zhang, X. Han, J. Deng, and W. Sun, "How to evaluate and remove the weakened bands in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502515.

[45] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[46] W. Zhirui, K. Yuzhuo, Z. Xuan, W. Yuelei, Z. Ting, and S. Xian, "SAR-aircraft-1.0: High resolution SAR aircraft detection and recognition dataset," *J. Radar*, vol. 12, no. 4, pp. 906–922, 2023.

[47] L. W. Sommer, T. Schuchert, and J. Beyerer, "Fast deep vehicle detection in aerial images," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2017, pp. 311–319.

[48] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. S. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1199–1208.

[49] J. Lai et al., "TSF: Transformer-based semantic filter for few-shot learning," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 1–19.

[50] H.-Y. Tseng, H.-Y. Lee, J.-B. Huang, and M.-H. Yang, "Cross-domain few-shot classification via learned feature-wise transformation," 2020, *arXiv:2001.08735*.

[51] H. Wang, H. Mai, Y. Gong, and Z.-H. Deng, "Towards well-generalizing meta-learning via adversarial task augmentation," *Artif. Intell.*, vol. 317, Apr. 2023, Art. no. 103875.

[52] W.-H. Li, X. Liu, and H. Bilen, "Universal representations: A unified look at multiple task and domain learning," *Int. J. Comput. Vis.*, vol. 132, no. 5, pp. 1521–1545, May 2024.

[53] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[54] P.-T. Jiang, C.-B. Zhang, Q. Hou, M.-M. Cheng, and Y. Wei, "LayerCAM: Exploring hierarchical class activation maps for localization," *IEEE Trans. Image Process.*, vol. 30, pp. 5875–5888, 2021.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.



**Jie Lai** received the M.S. degree in electronic information from Harbin Engineering University, Harbin, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include remote sensing image object detection and deep learning.



**Jie Geng** (Member, IEEE) received the B.S. degree in electronic information engineering and the Ph.D. degree in signal and information processing from Dalian University of Technology, Dalian, China, in 2013 and 2018, respectively.

He is currently an Associate Professor at the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include SAR image processing, remote sensing image classification, and few-shot learning.



**Weichen Ma** received the M.S. degree from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2025.

His research interests include SAR image processing and few-shot learning.



**Wen Jiang** (Member, IEEE) received the bachelor's and master's degrees from Information Engineering University, Zhengzhou, China, in 1994 and 1997, respectively, and the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2009.

She is currently a Professor at the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include information fusion, artificial intelligence, remote sensing image processing, and intelligent algorithm security.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:53:56 UTC from IEEE Xplore. Restrictions apply.